Supplementary Material

S.1. Sampling Strategy Results

Table 1: Performance on the full evaluation sets under direct uniform sampling.

Models	Datasets	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	0.7344	0.5627	0.5800	0.8000	0.0591	0.3896
	CG-Bench	0.7643	0.5701	0.6200	0.9150	0.0398	0.4141
	Video-MME	0.7487	0.5492	0.4600	0.8700	0.0420	0.4101
InternVL2.5-8B	LongVideoBench	0.8064	0.5620	0.9800	0.7500	0.0526	0.3377
	CG-Bench	0.8117	0.5772	1.0000	0.9650	0.0314	0.3324
	Video-MME	0.8075	0.5656	1.0000	0.9667	0.0484	0.3430
VideoLLaMA2-7B-16F	LongVideoBench	0.6845	0.5617	0.4300	0.6800	0.0579	0.5178
	CG-Bench	0.6867	0.5633	0.4550	0.8700	0.0435	0.5249
	Video-MME	0.6830	0.5566	0.5500	0.9100	0.0516	0.5274

Table 1 compares direct uniform sampling with the binwise configuration reported in Table 2 of the main text. Both strategies use $M = 16$ input frames, while all other generation and evaluation settings remain unchanged. Macro-averaged over the nine backbone–dataset combinations, binwise sampling improves Reward from 0.7475 to 0.7557, MV-Align from 0.5632 to 0.5670, CSR from 0.6750 to 0.7011, SPC from 0.8585 to 0.8867, and CIDEr-lite from 0.0474 to 0.0495, while reducing Rep from 0.4219 to 0.3908. Binwise sampling gives more favorable aggregate results and retains two separated observations within each coarse temporal region.

Complete Results of Ablation on Self-Reward Components

Full results for the ablation of the self-reward components are provided here in Table 2, with the corresponding macro-averaged results reported in Table 6 of the main text.

S.2. Human Evaluation

We recruit 36 volunteers, all holding a master degree or higher in electronic engineering, computer science, or a related field. We assign 12 volunteers to each model: LLaVA-Video-7B-Qwen2, InternVL2.5-8B, and VideoLLaMA2-7B-16F. Each group evaluates the same 60 videos, comprising 20 from each dataset. We conceal method

Table 2: Reward-variant ablation across different video-language backbones and datasets.

Model	Dataset	Reward variant	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	w/o Alignment	0.7081	0.4934	0.4400	0.7900	0.0604	0.3888
		w/o Readability	0.6968	0.4689	0.4300	0.7800	0.0634	0.4226
		w/o Rhythm	0.7040	0.4653	0.4500	0.7700	0.0601	0.3940
		w/o Control	0.7071	0.4659	0.4300	0.7950	0.0602	0.3991
		w/o Risk	0.7187	0.5057	0.4400	0.8050	0.0599	0.3899
	CG-Bench	w/o Alignment	0.7544	0.4958	0.7600	0.9300	0.0369	0.3900
		w/o Readability	0.7075	0.4923	0.7500	0.9250	0.0359	0.4140
		w/o Rhythm	0.7494	0.4912	0.7400	0.9300	0.0380	0.4156
		w/o Control	0.7544	0.4905	0.7300	0.9400	0.0371	0.4123
		w/o Risk	0.7654	0.5006	0.7600	0.9400	0.0374	0.3924
	Video-MME	w/o Alignment	0.7071	0.4624	0.4000	0.8150	0.0408	0.5103
		w/o Readability	0.6874	0.4684	0.4800	0.8050	0.0399	0.5041
		w/o Rhythm	0.7004	0.4693	0.4900	0.8100	0.0400	0.5103
		w/o Control	0.7145	0.4688	0.4700	0.8250	0.0409	0.5052
		w/o Risk	0.7478	0.5577	0.5000	0.8600	0.0411	0.4043
InternVL2.5-8B	LongVideoBench	w/o Alignment	0.7922	0.4561	0.9600	0.7000	0.0466	0.3464
		w/o Readability	0.7897	0.4768	0.9700	0.7500	0.0408	0.3328
		w/o Rhythm	0.7954	0.4689	0.9750	0.7000	0.0455	0.3449
		w/o Control	0.7923	0.4708	0.8800	0.6900	0.0456	0.3395
		w/o Risk	0.7954	0.4689	0.9700	0.7000	0.0455	0.3449
	CG-Bench	w/o Alignment	0.8012	0.4729	0.9300	0.9400	0.0304	0.3344
		w/o Readability	0.7955	0.4869	0.9400	0.9300	0.0303	0.3262
		w/o Rhythm	0.8026	0.4802	0.9550	0.9400	0.0313	0.3293
		w/o Control	0.8026	0.4832	0.9200	0.9300	0.0311	0.3293
		w/o Risk	0.8162	0.5802	1.0000	0.9600	0.0351	0.3239
	Video-MME	w/o Alignment	0.7955	0.4610	0.9250	0.9300	0.0371	0.3301
		w/o Readability	0.7903	0.4748	0.9000	0.9400	0.0367	0.3247
		w/o Rhythm	0.7966	0.4659	0.9350	0.9300	0.0365	0.3249
		w/o Control	0.7953	0.4661	0.9400	0.9300	0.0363	0.3253
		w/o Risk	0.8206	0.5659	1.0000	0.9500	0.0465	0.3133
VideoLLaMA2-7B-16F	LongVideoBench	w/o Alignment	0.6062	0.4385	0.4250	0.6900	0.0607	0.5276
		w/o Readability	0.6041	0.4409	0.4300	0.7000	0.0597	0.5094
		w/o Rhythm	0.5999	0.4403	0.4200	0.7400	0.0596	0.5179
		w/o Control	0.6081	0.4418	0.4300	0.6800	0.0603	0.5185
		w/o Risk	0.6865	0.5608	0.4500	0.7500	0.0640	0.5009
	CG-Bench	w/o Alignment	0.6076	0.4676	0.4300	0.9200	0.0409	0.5445
		w/o Readability	0.6073	0.4785	0.4450	0.9350	0.0400	0.5116
		w/o Rhythm	0.6034	0.4785	0.4200	0.9200	0.0438	0.5417
		w/o Control	0.5995	0.4790	0.4350	0.9300	0.0439	0.5107
		w/o Risk	0.6090	0.4761	0.4400	0.9350	0.0438	0.5230
	Video-MME	w/o Alignment	0.6101	0.4524	0.4550	0.8900	0.0455	0.4988
		w/o Readability	0.6032	0.4580	0.4600	0.8900	0.0472	0.4864
		w/o Rhythm	0.6057	0.4586	0.4400	0.8900	0.0453	0.4874
		w/o Control	0.6035	0.4567	0.4700	0.8700	0.0457	0.5071
		w/o Risk	0.7015	0.5568	0.5800	0.9300	0.0494	0.4848

labels and randomize script presentation order for blind assessment. After watching each video, volunteers rate the baseline and STARS scripts on a five-point scale (1 = poor, 2 = average, 3 = good, 4 = very good, and 5 = excellent) for visual grounding, temporal consistency, completeness, readability, and overall preference.

We average the 12 ratings per script and per dimension and use the means rounded to two decimal places as reported in the full tables, which are publicly available at `https://github.com/aoaoiiz/STARS/tree/main/HumanEvaluation`. Within each model–dataset combination, we compare STARS with the baseline using two-sided exact paired sign-flip permutation tests, treating videos as the paired units. We use Holm correction across the 45 dimension-specific comparisons. STARS achieves higher ratings in all comparisons, with adjusted $p < 0.001$.

We also examine the association between Reward and the equal-weight mean of the five rating dimensions using exploratory Spearman correlation analyses. For the 60 LLaVA-Video-7B-Qwen2 STARS outputs, we find a positive correlation ($\rho = 0.455$, adjusted $p = 0.0016$). This p-value incorporates the Holm correction for the six comparisons between model–method composite scores.

S.3. Comparison with a Proprietary Model Using Four Candidates

We conduct an additional experiment on LongVideoBench to account for the difference in candidate budgets between STARS and the proprietary models. GPT-5.6 Terra generates four candidate scripts per video, and the same reward criterion selects the final script. Table 3 shows the additional results with GPT generation using four candidates.

Table 3: Comparison of GPT-5.6 Terra and STARS on LongVideoBench. For $N = 4$, the final script is selected using the same reward criterion.

Model	N	Method	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
GPT-5.6 Terra	1	Single candidate	0.8150	0.4955	1.0000	0.8600	0.0412	0.2170
GPT-5.6 Terra	4	Reward selection	0.8374	0.5623	1.0000	0.8600	0.0505	0.2067
STARS (InternVL2.5-8B)	4	Reward selection	0.8154	0.5689	0.9800	0.7800	0.0555	0.3049

Using four candidates improves the GPT-5.6 Terra results in Reward, MV-Align, CIDEr-lite, and Rep, while CSR and SPC remain unchanged. The results therefore

show different strengths across the evaluated metrics. The additional GPT experiment used 51,752,893 input tokens and 17,428,315 output tokens, including reasoning tokens, totaling 69,181,208 tokens. STARS with open-source models can run locally without proprietary API calls or their associated charges. STARS's performance on selected metrics and its support for local deployment make it a practical option.

S.4. Statistical Robustness Across Random Seeds

We present the experimental results of the baseline and STARS obtained with random seeds 42, 123, and 456 on the same evaluation videos. Tables 4–6 give the individual results, and Table 7 reports the mean and sample standard deviation across the three runs. We calculate 95% confidence intervals based on the t distribution and use two-sided paired t tests to compare STARS with the baseline, pairing results by seed ($n = 3$, $df = 2$). For the summaries by model, dataset, and overall, scores are first averaged within each seed with equal weights. The intervals and tests are then calculated across the three seed averages. Differences are defined as STARS minus baseline. Tables 8–11 report the confidence intervals and raw p values.

STARS achieves higher scores on five metrics and lower Rep for every combination of model and dataset in all three runs. The overall mean Reward increases from 0.7163 to 0.7557, while Rep decreases from 0.4185 to 0.3908. The 95% confidence intervals for the overall mean paired differences exclude zero for all six metrics (all raw $p < 0.05$). These results reflect variation across random seeds and support the statistical robustness of STARS under the evaluated settings.

Table 4: Baseline and STARS results with random seed 42.

Model	Dataset	Method	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	Baseline	0.6625	0.5593	0.5300	0.7900	0.0616	0.3952
		Ours	0.7327	0.5688	0.5900	0.8800	0.0699	0.3899
	CG-Bench	Baseline	0.7160	0.5729	0.4500	0.9300	0.0384	0.4122
		Ours	0.7883	0.5771	0.7800	0.9500	0.0385	0.3675
	Video-MME	Baseline	0.6859	0.5540	0.3400	0.8600	0.0408	0.4225
		Ours	0.7488	0.5591	0.5000	0.8700	0.0410	0.4002
InternVL2.5-8B	LongVideoBench	Baseline	0.8059	0.5663	0.8100	0.7100	0.0511	0.3328
		Ours	0.8160	0.5678	0.9800	0.7900	0.0559	0.3004
	CG-Bench	Baseline	0.8171	0.5706	0.9200	0.9500	0.0341	0.3312
		Ours	0.8178	0.5822	1.0000	0.9600	0.0352	0.3216
	Video-MME	Baseline	0.7927	0.5602	0.9200	0.9450	0.0455	0.3228
		Ours	0.8208	0.5649	1.0000	0.9600	0.0466	0.3110
VideoLLaMA2-7B-16F	LongVideoBench	Baseline	0.6323	0.5598	0.3200	0.6800	0.0625	0.5324
		Ours	0.6874	0.5608	0.4500	0.7700	0.0637	0.4949
	CG-Bench	Baseline	0.6480	0.5706	0.3600	0.9100	0.0445	0.5220
		Ours	0.6879	0.5766	0.4900	0.9700	0.0451	0.4166
	Video-MME	Baseline	0.6963	0.5510	0.5100	0.9350	0.0447	0.5114
		Ours	0.7023	0.5572	0.5800	0.9500	0.0499	0.4840

Table 5: Baseline and STARS results with random seed 123.

Model	Dataset	Method	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	Baseline	0.6626	0.5612	0.5400	0.8000	0.0614	0.3945
		Ours	0.7581	0.5679	0.5500	0.8700	0.0698	0.3900
	CG-Bench	Baseline	0.7164	0.5728	0.4000	0.9400	0.0383	0.4121
		Ours	0.7884	0.5745	0.7500	0.9800	0.0402	0.3852
	Video-MME	Baseline	0.6860	0.5541	0.3300	0.8500	0.0407	0.4224
		Ours	0.7482	0.5593	0.5000	0.8600	0.0412	0.3998
InternVL2.5-8B	LongVideoBench	Baseline	0.8035	0.5676	0.7700	0.7600	0.0510	0.3329
		Ours	0.8168	0.5717	0.9800	0.7700	0.0561	0.2986
	CG-Bench	Baseline	0.8168	0.5706	0.9400	0.9600	0.0343	0.3310
		Ours	0.8169	0.5821	1.0000	0.9700	0.0352	0.3217
	Video-MME	Baseline	0.7928	0.5591	0.8400	0.9350	0.0450	0.3219
		Ours	0.8209	0.5655	1.0000	0.9400	0.0468	0.3108
VideoLLaMA2-7B-16F	LongVideoBench	Baseline	0.6296	0.5369	0.3100	0.7000	0.0625	0.5316
		Ours	0.6867	0.5608	0.4300	0.7450	0.0639	0.4947
	CG-Bench	Baseline	0.6465	0.5712	0.4000	0.9200	0.0441	0.5225
		Ours	0.6860	0.5782	0.4600	0.9300	0.0450	0.4167
	Video-MME	Baseline	0.6961	0.5505	0.5600	0.8700	0.0444	0.5113
		Ours	0.7024	0.5573	0.5900	0.9100	0.0498	0.4842

Table 6: Baseline and STARS results with random seed 456.

Model	Dataset	Method	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	Baseline	0.6462	0.5595	0.5200	0.7500	0.0612	0.3941
		Ours	0.7253	0.5604	0.5400	0.8000	0.0700	0.3898
	CG-Bench	Baseline	0.7243	0.5529	0.4100	0.9200	0.0364	0.4114
		Ours	0.7795	0.5602	0.7500	0.9500	0.0389	0.3645
	Video-MME	Baseline	0.6837	0.5530	0.3200	0.8400	0.0409	0.4233
		Ours	0.7464	0.5547	0.5000	0.8500	0.0411	0.4129
InternVL2.5-8B	LongVideoBench	Baseline	0.7987	0.5665	0.7600	0.7500	0.0513	0.3272
		Ours	0.8134	0.5672	0.9800	0.7800	0.0545	0.3157
	CG-Bench	Baseline	0.8135	0.5721	0.9600	0.9400	0.0327	0.3296
		Ours	0.8139	0.5764	1.0000	0.9500	0.0350	0.3285
	Video-MME	Baseline	0.8046	0.5556	0.8800	0.9400	0.0457	0.3202
		Ours	0.8201	0.5673	1.0000	0.9500	0.0461	0.3181
VideoLLaMA2-7B-16F	LongVideoBench	Baseline	0.6320	0.5602	0.3300	0.7200	0.0623	0.5269
		Ours	0.6854	0.5608	0.4700	0.7350	0.0644	0.5132
	CG-Bench	Baseline	0.6387	0.5700	0.3800	0.9000	0.0440	0.5056
		Ours	0.6931	0.5735	0.4900	0.9200	0.0442	0.4358
	Video-MME	Baseline	0.6923	0.5512	0.5500	0.9250	0.0459	0.4985
		Ours	0.6998	0.5559	0.5700	0.9300	0.0486	0.4862

Table 7: Mean ± sample standard deviation across three runs with seeds 42, 123, and 456.

Model	Dataset	Method	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
LLaVA-Video-7B-Qwen2	LongVideoBench	Baseline	0.6571 ± 0.009440	0.5600 ± 0.001044	0.5300 ± 0.010000	0.7800 ± 0.026458	0.0614 ± 0.000200	0.3946 ± 0.000557
		Ours	0.7387 ± 0.017203	0.5657 ± 0.004612	0.5600 ± 0.026458	0.8500 ± 0.043589	0.0699 ± 0.000100	0.3899 ± 0.000100
	CG-Bench	Baseline	0.7189 ± 0.004681	0.5662 ± 0.011518	0.4200 ± 0.026458	0.9300 ± 0.010000	0.0377 ± 0.001127	0.4119 ± 0.000436
		Ours	0.7854 ± 0.005110	0.5706 ± 0.009100	0.7600 ± 0.017321	0.9600 ± 0.017321	0.0392 ± 0.000889	0.3724 ± 0.011186
	Video-MME	Baseline	0.6852 ± 0.001300	0.5537 ± 0.000608	0.3300 ± 0.010000	0.8500 ± 0.010000	0.0408 ± 0.000100	0.4227 ± 0.000493
		Ours	0.7478 ± 0.001249	0.5577 ± 0.002600	0.5000 ± 0.000000	0.8600 ± 0.010000	0.0411 ± 0.000100	0.4043 ± 0.007451
InternVL2.5-8B	LongVideoBench	Baseline	0.8027 ± 0.003666	0.5668 ± 0.000700	0.7800 ± 0.026458	0.7400 ± 0.026458	0.0511 ± 0.000153	0.3310 ± 0.003262
		Ours	0.8154 ± 0.001778	0.5689 ± 0.002443	0.9800 ± 0.000000	0.7800 ± 0.010000	0.0555 ± 0.000872	0.3049 ± 0.009396
	CG-Bench	Baseline	0.8158 ± 0.001997	0.5711 ± 0.000866	0.9400 ± 0.020000	0.9500 ± 0.010000	0.0337 ± 0.000872	0.3306 ± 0.000872
		Ours	0.8162 ± 0.002042	0.5802 ± 0.003320	1.0000 ± 0.000000	0.9600 ± 0.010000	0.0351 ± 0.000115	0.3239 ± 0.003955
	Video-MME	Baseline	0.7967 ± 0.006842	0.5583 ± 0.002402	0.8800 ± 0.040000	0.9400 ± 0.005000	0.0454 ± 0.000361	0.3216 ± 0.001320
		Ours	0.8206 ± 0.000436	0.5659 ± 0.001249	1.0000 ± 0.000000	0.9500 ± 0.010000	0.0465 ± 0.000361	0.3133 ± 0.004158
VideoLLaMA2-7B-16F	LongVideoBench	Baseline	0.6313 ± 0.001480	0.5523 ± 0.013338	0.3200 ± 0.010000	0.7000 ± 0.020000	0.0624 ± 0.000115	0.5303 ± 0.002972
		Ours	0.6865 ± 0.001015	0.5608 ± 0.000000	0.4500 ± 0.020000	0.7500 ± 0.018028	0.0640 ± 0.000361	0.5009 ± 0.010624
	CG-Bench	Baseline	0.6444 ± 0.004993	0.5706 ± 0.000600	0.3800 ± 0.020000	0.9100 ± 0.010000	0.0442 ± 0.000265	0.5167 ± 0.009616
		Ours	0.6890 ± 0.003676	0.5761 ± 0.002390	0.4800 ± 0.017321	0.9400 ± 0.026458	0.0448 ± 0.000493	0.4230 ± 0.011056
	Video-MME	Baseline	0.6949 ± 0.002254	0.5509 ± 0.000361	0.5400 ± 0.026458	0.9100 ± 0.035000	0.0450 ± 0.000794	0.5071 ± 0.007419
		Ours	0.7015 ± 0.001473	0.5568 ± 0.000781	0.5800 ± 0.010000	0.9300 ± 0.020000	0.0494 ± 0.000723	0.4848 ± 0.001217

Table 8: 95% confidence intervals and paired comparisons for each model and dataset. Δ = STARS − Baseline. Bold values indicate $p < 0.05$.

Model	Dataset	Metric	Baseline 95% CI	STARS 95% CI	Paired Δ 95% CI	Raw p paired t test
LLaVA-Video-7B-Qwen2	LongVideoBench	Reward	[0.633650, 0.680550]	[0.695964, 0.781436]	[0.049719, 0.113481]	**0.00814493**
		MV-Align	[0.557406, 0.562594]	[0.554243, 0.577157]	[−0.005196, 0.016596]	0.15326819
		CSR	[0.505159, 0.554841]	[0.494276, 0.625724]	[−0.035724, 0.095724]	0.18849733
		SPC	[0.714276, 0.845724]	[0.741719, 0.958281]	[0.020317, 0.119683]	**0.02614832**
		CIDEr-lite	[0.060903, 0.061897]	[0.069652, 0.070148]	[0.007843, 0.009157]	**0.00032280**
		Rep	[0.393217, 0.395983]	[0.389652, 0.390148]	[−0.006014, −0.003386]	**0.00419855**
	CG-Bench	Reward	[0.707272, 0.730528]	[0.772707, 0.798093]	[0.042187, 0.090813]	**0.00714304**
		MV-Align	[0.537587, 0.594813]	[0.547994, 0.593206]	[−0.002569, 0.011369]	0.11299456
		CSR	[0.354276, 0.485724]	[0.716973, 0.803027]	[0.315159, 0.364841]	**0.00028823**
		SPC	[0.905159, 0.954841]	[0.916973, 1.003027]	[0.005159, 0.054841]	**0.03509872**
		CIDEr-lite	[0.034901, 0.040499]	[0.036992, 0.041408]	[−0.001603, 0.004603]	0.17302323
		Rep	[0.410817, 0.412983]	[0.344612, 0.400188]	[−0.066744, −0.012256]	**0.02474685**
	Video-MME	Reward	[0.681971, 0.688429]	[0.744697, 0.750903]	[0.061704, 0.063496]	**0.00001106**
		MV-Align	[0.552189, 0.555211]	[0.551241, 0.564159]	[−0.000950, 0.008950]	0.07368353
		CSR	[0.305159, 0.354841]	-	[0.145159, 0.194841]	**0.00115141**
		SPC	[0.825159, 0.874841]	[0.835159, 0.884841]	-	-
		CIDEr-lite	[0.040552, 0.041048]	[0.040852, 0.041348]	[−0.000130, 0.000730]	0.09546597
		Rep	[0.421508, 0.423959]	[0.385792, 0.422808]	[−0.035720, −0.001147]	**0.04436610**
InternVL2.5-8B	LongVideoBench	Reward	[0.793593, 0.811807]	[0.810984, 0.819816]	[0.006842, 0.018558]	**0.01129635**
		MV-Align	[0.565061, 0.568539]	[0.562830, 0.574970]	[−0.002316, 0.006516]	0.17736622
		CSR	[0.714276, 0.845724]	-	[0.134276, 0.265724]	**0.00578278**
		SPC	[0.674276, 0.805724]	[0.755159, 0.804841]	[−0.049567, 0.129567]	0.19461273
		CIDEr-lite	[0.050754, 0.051513]	[0.053334, 0.057666]	[0.001829, 0.006904]	**0.01775474**
		Rep	[0.322862, 0.339071]	[0.281558, 0.328242]	[−0.057493, 0.005360]	0.07033135
	CG-Bench	Reward	[0.810838, 0.820762]	[0.811127, 0.821273]	[−0.000345, 0.001145]	0.14719713
		MV-Align	[0.568949, 0.573251]	[0.571986, 0.588481]	[−0.001265, 0.019532]	0.06343257
		CSR	[0.890317, 0.989683]	-	[0.010317, 0.109683]	**0.03509872**
		SPC	[0.925159, 0.974841]	[0.935159, 0.984841]	-	-
		CIDEr-lite	[0.031534, 0.035866]	[0.034846, 0.035420]	[−0.000448, 0.003314]	0.08177494
		Rep	[0.328434, 0.332766]	[0.314108, 0.333759]	[−0.018648, 0.005315]	0.13900214
	Video-MME	Reward	[0.779704, 0.813696]	[0.819517, 0.821683]	[0.005829, 0.041971]	**0.02952114**
		MV-Align	[0.552333, 0.564267]	[0.562797, 0.569003]	[−0.001470, 0.016670]	0.06905425
		CSR	[0.780634, 0.979366]	-	[0.020634, 0.219366]	**0.03509872**
		SPC	[0.927579, 0.952421]	[0.925159, 0.974841]	[−0.002421, 0.022421]	0.07417990
		CIDEr-lite	[0.044504, 0.046296]	[0.045604, 0.047396]	[−0.000639, 0.002839]	0.11263387
		Rep	[0.318353, 0.324913]	[0.302971, 0.323629]	[−0.021771, 0.005105]	0.11643676
VideoLLaMA2-7B-16F	LongVideoBench	Reward	[0.627624, 0.634976]	[0.683979, 0.689021]	[0.050599, 0.059801]	**0.00037502**
		MV-Align	[0.519166, 0.585434]	-	[−0.024634, 0.041634]	0.38473068
		CSR	[0.295159, 0.344841]	[0.400317, 0.499683]	[0.105159, 0.154841]	**0.00196657**
		SPC	[0.650317, 0.749683]	[0.705217, 0.794783]	[−0.043774, 0.143774]	0.14874347
		CIDEr-lite	[0.062146, 0.062720]	[0.063104, 0.064896]	[0.000393, 0.002741]	**0.02901680**
		Rep	[0.522918, 0.537682]	[0.474543, 0.527324]	[−0.063079, 0.004346]	0.06438726
	CG-Bench	Reward	[0.631997, 0.656803]	[0.679869, 0.698131]	[0.023511, 0.065689]	**0.01186265**
		MV-Align	[0.569110, 0.572090]	[0.570164, 0.582036]	[0.001022, 0.009978]	**0.03399692**
		CSR	[0.330317, 0.429683]	[0.436973, 0.523027]	[0.010433, 0.189567]	**0.04070578**
		SPC	[0.885159, 0.934841]	[0.874276, 1.005724]	[−0.035724, 0.095724]	0.18849733
		CIDEr-lite	[0.043543, 0.044857]	[0.043541, 0.045992]	[−0.000306, 0.001439]	0.10773140
		Rep	[0.492812, 0.540588]	[0.395568, 0.450499]	[−0.145014, −0.042319]	**0.01584797**
	Video-MME	Reward	[0.689301, 0.700499]	[0.697841, 0.705159]	[0.004628, 0.008572]	**0.00478635**
		MV-Align	[0.550004, 0.551796]	[0.554860, 0.558740]	[0.003213, 0.008587]	**0.01101884**
		CSR	[0.474276, 0.605724]	[0.555159, 0.604841]	[−0.025724, 0.105724]	0.12011731
		SPC	[0.823055, 0.996945]	[0.880317, 0.979683]	[−0.024783, 0.064783]	0.19461273
		CIDEr-lite	[0.043028, 0.046972]	[0.047636, 0.051230]	[0.000696, 0.008171]	**0.03630779**
		Rep	[0.488637, 0.525497]	[0.481778, 0.487822]	[−0.043711, −0.000822]	**0.04662636**

Intervals are based on the t distribution with $n = 3$ and $df = 2$. All p values are from two-sided paired t tests. "-" marks intervals or tests omitted because the estimated standard error was zero. Intervals are not truncated to [0, 1].

Table 9: 95% confidence intervals and paired comparisons by model, averaged equally across the three datasets within each seed. Δ = STARS − Baseline. Bold values indicate $p < 0.05$.

Model	Metric	Baseline 95% CI	STARS 95% CI	Paired Δ 95% CI	Raw p paired t test
LLaVA-Video-7B-Qwen2	Reward	[0.682041, 0.692093]	[0.739227, 0.775373]	[0.056171, 0.084295]	**0.00215835**
	MV-Align	[0.549539, 0.570394]	[0.551187, 0.578146]	[0.000998, 0.008402]	**0.03192001**
	CSR	[0.396811, 0.456522]	[0.570573, 0.642760]	[0.171720, 0.188280]	**0.00011429**
	SPC	[0.817240, 0.889427]	[0.839632, 0.940368]	[0.022324, 0.051009]	**0.00816340**
	CIDEr-lite	[0.045616, 0.047651]	[0.049308, 0.050826]	[0.002180, 0.004686]	**0.00711841**
	Rep	[0.409259, 0.410230]	[0.381650, 0.396084]	[−0.028491, −0.013265]	**0.00710553**
InternVL2.5-8B	Reward	[0.803493, 0.806640]	[0.813958, 0.820842]	[0.007619, 0.017047]	**0.00779921**
	MV-Align	[0.563963, 0.566837]	[0.568199, 0.575157]	[0.003962, 0.008594]	**0.00727208**
	CSR	[0.825264, 0.908069]	-	[0.085264, 0.168069]	**0.00572153**
	SPC	[0.855966, 0.897368]	[0.882324, 0.911009]	[−0.013889, 0.053889]	0.12635769
	CIDEr-lite	[0.042994, 0.043828]	[0.044599, 0.046823]	[0.001510, 0.003090]	**0.00631096**
	Rep	[0.323268, 0.332199]	[0.299561, 0.328528]	[−0.032600, 0.005223]	0.08947491
VideoLLaMA2-7B-16F	Reward	[0.651120, 0.662613]	[0.690940, 0.693726]	[0.029036, 0.041897]	**0.00177105**
	MV-Align	[0.547033, 0.568833]	[0.561960, 0.567173]	[−0.006261, 0.019527]	0.15730653
	CSR	[0.377240, 0.449427]	[0.481425, 0.525241]	[0.040317, 0.139683]	**0.01606530**
	SPC	[0.816948, 0.863052]	[0.823136, 0.923531]	[−0.018544, 0.085210]	0.10971781
	CIDEr-lite	[0.050045, 0.051044]	[0.052016, 0.053450]	[0.001029, 0.003349]	**0.01483990**
	Rep	[0.501480, 0.534564]	[0.450633, 0.488545]	[−0.083931, −0.012936]	**0.02781101**

Intervals are based on the t distribution with $n = 3$ and $df = 2$. All p values are from two-sided paired t tests. "-" marks intervals or tests omitted because the estimated standard error was zero. Intervals are not truncated to [0, 1].

Table 10: 95% confidence intervals and paired comparisons by dataset, averaged equally across the three models within each seed. Δ = STARS − Baseline. Bold values indicate $p < 0.05$.

Dataset	Metric	Baseline 95% CI	STARS 95% CI	Paired Δ 95% CI	Raw p paired t test
LongVideoBench	Reward	[0.686642, 0.707425]	[0.731009, 0.762724]	[0.037098, 0.062568]	**0.00350907**
	MV-Align	[0.550085, 0.569315]	[0.559962, 0.570304]	[−0.008371, 0.019238]	0.23244312
	CSR	[0.521425, 0.565241]	[0.638492, 0.688175]	[0.103439, 0.136561]	**0.00102722**
	SPC	[0.706878, 0.773122]	[0.741456, 0.845210]	[−0.019445, 0.126112]	0.08757448
	CIDEr-lite	[0.058150, 0.058495]	[0.062754, 0.063513]	[0.004466, 0.005156]	**0.00027723**
	Rep	[0.413094, 0.424151]	[0.382089, 0.415066]	[−0.042013, 0.001924]	0.05918241
CG-Bench	Reward	[0.724414, 0.728319]	[0.760388, 0.766679]	[0.035964, 0.038369]	**0.00005657**
	MV-Align	[0.560047, 0.578553]	[0.563565, 0.587724]	[0.003447, 0.009242]	**0.01107859**
	CSR	[0.571720, 0.588280]	[0.721825, 0.771508]	[0.136811, 0.196522]	**0.00172884**
	SPC	[0.905159, 0.954841]	[0.924649, 0.982018]	[0.008991, 0.037676]	**0.01980394**
	CIDEr-lite	[0.036736, 0.040330]	[0.038724, 0.040676]	[−0.000166, 0.002499]	0.06380517
	Rep	[0.410697, 0.428769]	[0.363088, 0.383156]	[−0.064028, −0.029194]	**0.00745789**
Video-MME	Reward	[0.722875, 0.728325]	[0.754046, 0.759220]	[0.025724, 0.036343]	**0.00157724**
	MV-Align	[0.551992, 0.556608]	[0.558302, 0.561964]	[0.004751, 0.006916]	**0.00185604**
	CSR	[0.566772, 0.599894]	[0.685053, 0.701614]	[0.088092, 0.131908]	**0.00213578**
	SPC	[0.864626, 0.935374]	[0.883478, 0.943189]	[0.000913, 0.025754]	**0.04381711**
	CIDEr-lite	[0.042729, 0.044737]	[0.044785, 0.046571]	[0.000061, 0.003828]	**0.04712416**
	Rep	[0.410364, 0.423924]	[0.390185, 0.411415]	[−0.033725, 0.001036]	0.05599928

Intervals are based on the t distribution with $n = 3$ and $df = 2$. All p values are from two-sided paired t tests. Intervals are not truncated to $[0, 1]$.

Table 11: 95% confidence intervals and paired comparisons for the overall results, averaged equally across all nine model and dataset combinations within each seed. Δ = STARS − Baseline. Bold values indicate $p < 0.05$.

Metric	Baseline 95% CI	STARS 95% CI	Paired Δ 95% CI	Raw p paired t test
Reward	[0.713103, 0.719564]	[0.749119, 0.762237]	[0.034552, 0.044137]	**0.00080040**
MV-Align	[0.557538, 0.564662]	[0.560664, 0.573277]	[0.000590, 0.011151]	**0.04103519**
CSR	[0.558937, 0.578841]	[0.685743, 0.716479]	[0.126702, 0.137743]	**0.00009414**
SPC	[0.849766, 0.863567]	[0.854213, 0.919120]	[0.000144, 0.059856]	**0.04955499**
CIDEr-lite	[0.046476, 0.047250]	[0.048783, 0.050224]	[0.002035, 0.003247]	**0.00283426**
Rep	[0.411394, 0.425606]	[0.379347, 0.402319]	[−0.046214, −0.009120]	**0.02342583**

Intervals are based on the t distribution with $n = 3$ and $df = 2$. All p values are from two-sided paired t tests. Intervals are not truncated to $[0, 1]$.

S.5. Sensitivity to Reward Weights

We examined the sensitivity of STARS to six alternative reward weight configurations. These included equal weighting and five configurations in which one component weight was multiplied by 0.5, followed by renormalization of all weights. Equal weighting assigned a weight of 0.2 to each of the five active components: alignment, readability, rhythm, control, and risk. For each video and backbone, all configurations reuse the same four candidate scripts with seed 42. Candidates are selected using the weights of the corresponding configuration. The reported Reward was evaluated using the default weights for every selected script, providing a common scale for comparison.

Table 12 summarizes the results across the three models and three datasets. Every alternative configuration outperformed the baseline on all six averaged metrics. Across these configurations, CSR ranged from 0.6800 to 0.6956, compared with 0.5733 for the baseline. Rep ranged from 0.3892 to 0.4123, remaining below the baseline value of 0.4203. The first STARS row uses the default weights and achieves the best averaged performance. These results show that STARS retained its overall advantage under equal weighting and all five individual weight perturbations.

Table 12: Sensitivity to reward weights. Values are equally weighted averages over the nine combinations of three models and three datasets.

Method / reward weights	Reward	MV-Align	CSR	SPC	CIDEr-lite	Rep
Baseline	0.7174	0.5627	0.5733	0.8567	0.0470	0.4203
STARS (0.352900, 0.176500, 0.176500, 0.235300, 0.058800)	0.7558	0.5683	0.7078	0.9000	0.0495	0.3873
STARS (0.200000, 0.200000, 0.200000, 0.200000, 0.200000)	0.7539	0.5648	0.6856	0.8694	0.0482	0.4024
STARS (0.214255, 0.214316, 0.214316, 0.285714, 0.071398)	0.7539	0.5652	0.6844	0.8639	0.0482	0.4090
STARS (0.387058, 0.096792, 0.193584, 0.258075, 0.064491)	0.7540	0.5650	0.6811	0.8722	0.0482	0.4098
STARS (0.387058, 0.193584, 0.096792, 0.258075, 0.064491)	0.7537	0.5674	0.6800	0.8717	0.0493	0.4123
STARS (0.399955, 0.200034, 0.200034, 0.133337, 0.066640)	0.7544	0.5671	0.6833	0.8678	0.0493	0.4096
STARS (0.363590, 0.181846, 0.181846, 0.242427, 0.030291)	0.7555	0.5671	0.6956	0.8694	0.0495	0.3892